Day 073 | Unit 05 | Teacher plan

Represent and Solve Equations and Inequalities

Connect Rules, Tables, and Graphs

Standards: 6.EE.C.9

Learning target: I can connect an equation, a table, and a graph.

enVision Topic 4: Represent and Solve Equations and Inequalities | Spectrum companion Chapter 5

0–5 minutes / Launch

Read the target. Ask for an estimate or a sketch before calculating. Connect the new idea to the previous lesson.

5–17 / Model

Use Worked Example 1, then Example 2. Ask students to name the next step and explain a choice. Choose input values, substitute to find outputs, and write ordered pairs. Label horizontal input and vertical output axes. Plot each pair and interpret a point in the situation.

17–42 / Practice

Assign questions 1–10 in order. Use 1–3 for foundations, 4–7 for application, and 8–10 for state-format response and reasoning. The example button provides a separate question with a worked method.

42–50 / Discuss and check

Compare representations. Review a selected answer and a plausible error. Use question 10 as an exit discussion; ask students to check units, scale, or substitution.

Support and extension

Offer number lines, fraction strips, grid paper, or a table template as appropriate. Read multi-part directions one part at a time. For extension, ask for another representation or a counterexample.

Companion and state alignment

Use the matching skill in the edition you own. No unverified publisher page numbers are assigned. AASA references below identify related public sample items by standard and format; all course questions are original.

AASA sample item references: 5

Day 073 | Unit 05 | Lesson

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01 / Learn

Choose input values, substitute to find outputs, and write ordered pairs. Label horizontal input and vertical output axes. Plot each pair and interpret a point in the situation.

Worked example 1

For y = 7x, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes.

1. Choose input values, substitute to find outputs, and write ordered pairs.

2. Label horizontal input and vertical output axes.

3. Substitute each input into y = 7x

4. x is horizontal and y vertical.

5. State the result in the form requested: (0, 0); (1, 7); (2, 14); (3, 21).

Answer: (0, 0); (1, 7); (2, 14); (3, 21)

Worked example 2

A machine uses 6 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.

1. Choose input values, substitute to find outputs, and write ordered pairs.

2. Hours are the input.

3. Multiply each hour value by 6 to obtain liters.

4. Graph hours horizontally and liters vertically.

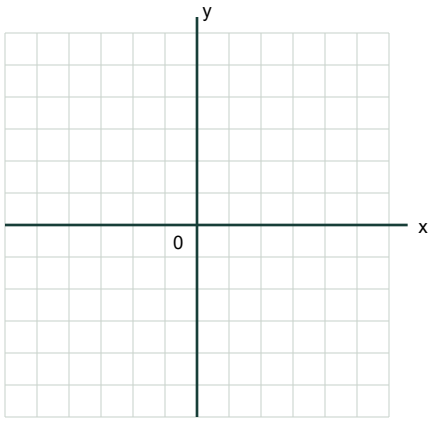
Answer: (0,0), (1,6), (2,12); independent h; L = 6h

Day 073 | Your turn

Work in order: foundation → application → explanation. Show calculations and label units.

01 / Constructed response

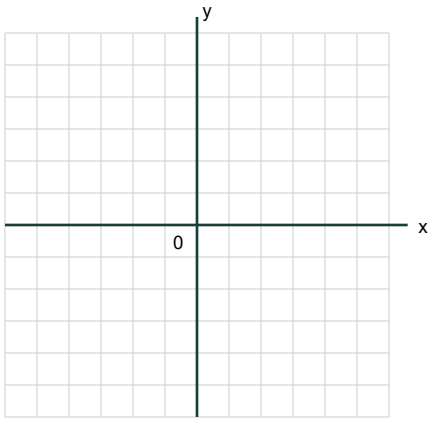
For y = 2x, make a table for x = 0, 1, 2, 3. Plot the ordered pairs and label both axes.



Label the scale and units.

02 / Constructed response

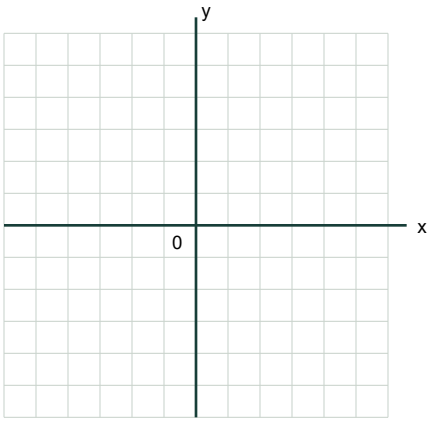
For y = 2x, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes.



Label the scale and units.

03 / Constructed response

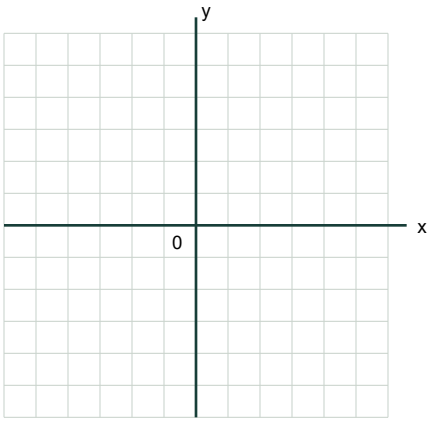
For y = 6x, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes.



Label the scale and units.

04 / Constructed response

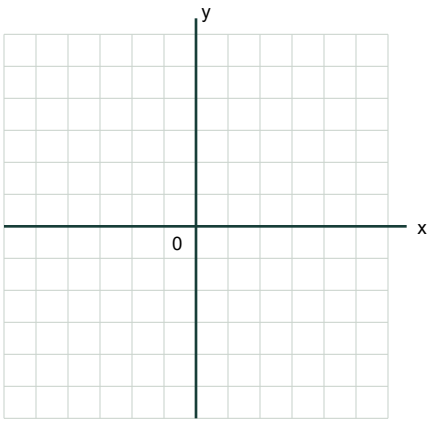
For y = 6x + 3, make a table for x = 0, 1, 2, 4. Plot the ordered pairs and label both axes.



Label the scale and units.

05 / Constructed response

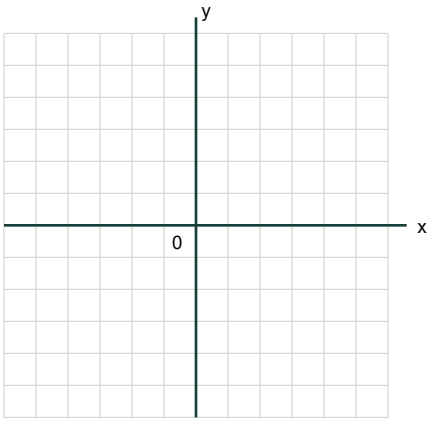
For y = 2x + 5, make a table for x = 0, 1, 2, 5. Plot the ordered pairs and label both axes.



Label the scale and units.

06 / Constructed response

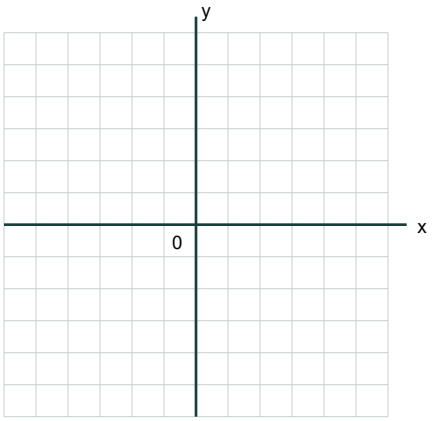
For y = 8x + 7, make a table for x = 0, 1, 2, 4. Plot the ordered pairs and label both axes.



Label the scale and units.

07 / Constructed response

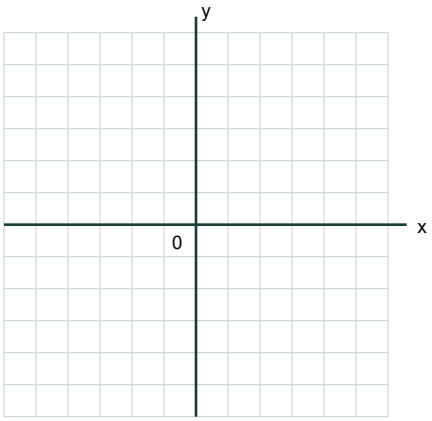
For y = 5x + 4, make a table for x = 0, 1, 2, 2. Plot the ordered pairs and label both axes. Explain what happens to y each time x increases by 1.



Label the scale and units.

08 / Table completion, equation, and graph

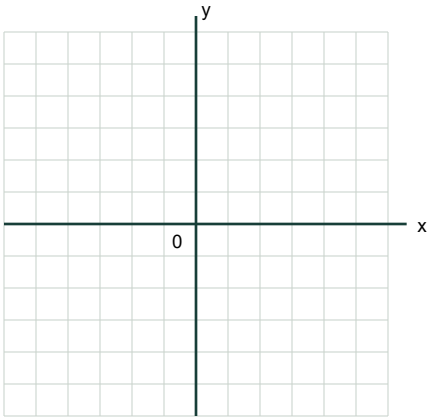
A machine uses 2 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (2, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

09 / Table completion, equation, and graph

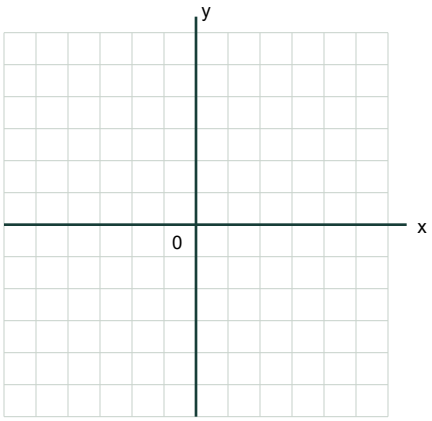
A machine uses 4 liters per hour. Part A: complete the (hours, liters) table: (0, __), (1, __), (4, __). Part B: choose the independent variable and write an equation for L in terms of h. Plot your pairs.



Label the scale and units.

10 / Constructed response

For y = 8x + 2, make a table for x = 0, 1, 2, 6. Plot the ordered pairs and label both axes. Explain what happens to y each time x increases by 1.



Label the scale and units.

Day 073 | Unit 05 | Answer key

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1. (0, 0); (1, 2); (2, 4); (3, 6)

Substitute each input into y = 2x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

2. (0, 0); (1, 2); (2, 4); (6, 12)

Substitute each input into y = 2x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

3. (0, 0); (1, 6); (2, 12); (6, 36)

Substitute each input into y = 6x; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

4. (0, 3); (1, 9); (2, 15); (4, 27)

Substitute each input into y = 6x + 3; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

5. (0, 5); (1, 7); (2, 9); (5, 15)

Substitute each input into y = 2x + 5; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

6. (0, 7); (1, 15); (2, 23); (4, 39)

Substitute each input into y = 8x + 7; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

7. (0, 4); (1, 9); (2, 14); y increases by 5

Substitute each input into y = 5x + 4; x is horizontal and y vertical.

6.EE.C.9 | Constructed response

8. (0,0), (1,2), (2,4); independent h; L = 2h

Hours are the input. Multiply each hour value by 2 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

9. (0,0), (1,4), (4,16); independent h; L = 4h

Hours are the input. Multiply each hour value by 4 to obtain liters. Graph hours horizontally and liters vertically.

6.EE.C.9 | Table completion, equation, and graph

10. (0, 2); (1, 10); (2, 18); (6, 50); y increases by 8

Substitute each input into y = 8x + 2; x is horizontal and y vertical.

6.EE.C.9 | Constructed response